

LECTURE

Intro to Modules





What are Modules?

A module is a container for related resources that are used together. They are the primary way to package and reuse configurations.



Improved Organization

Break large configurations into logical units to easily reuse over and over again.



Easier Collaboration

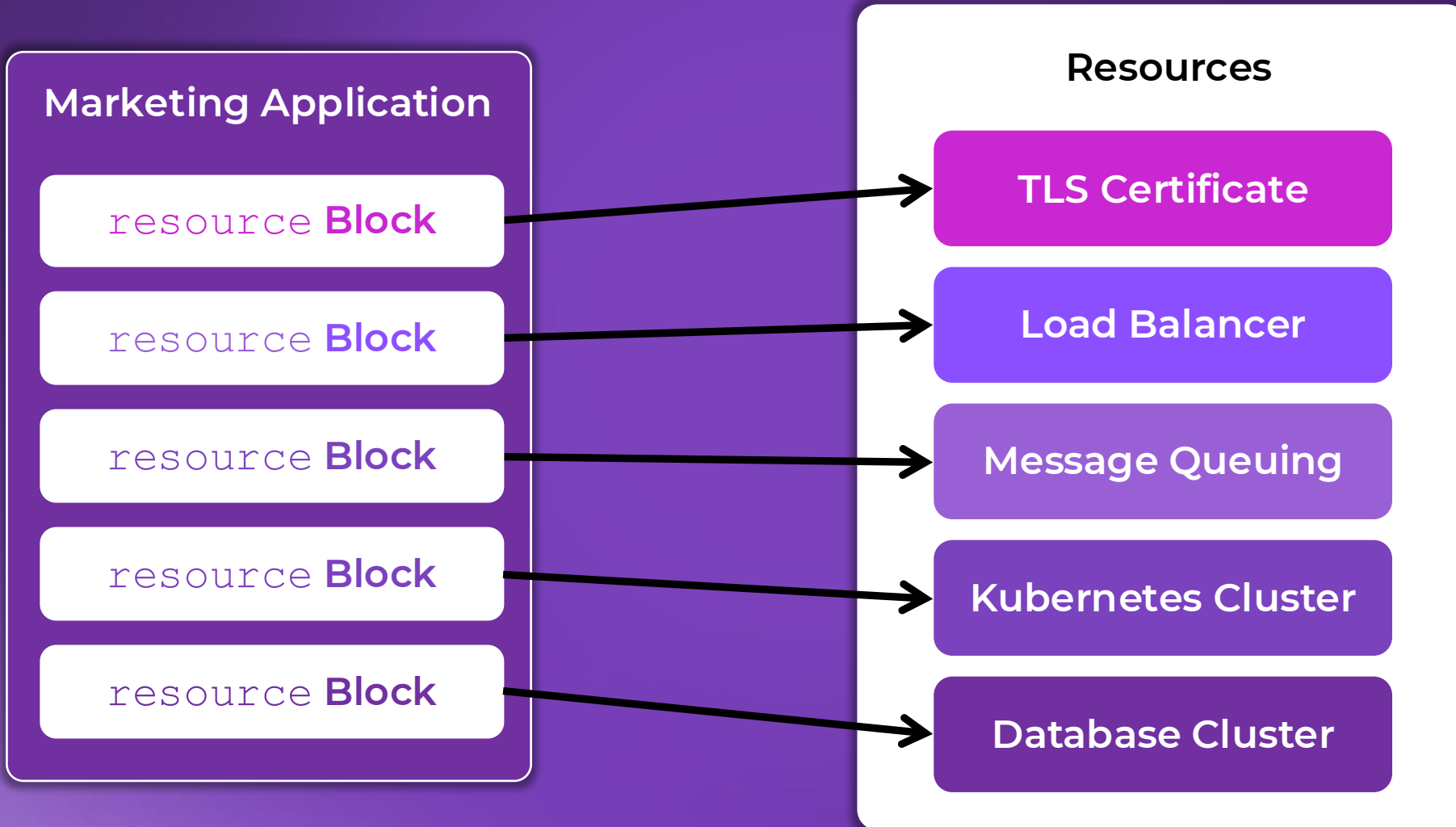
Teams can share and maintain standardized building blocks or publish approved modules for organizational use.



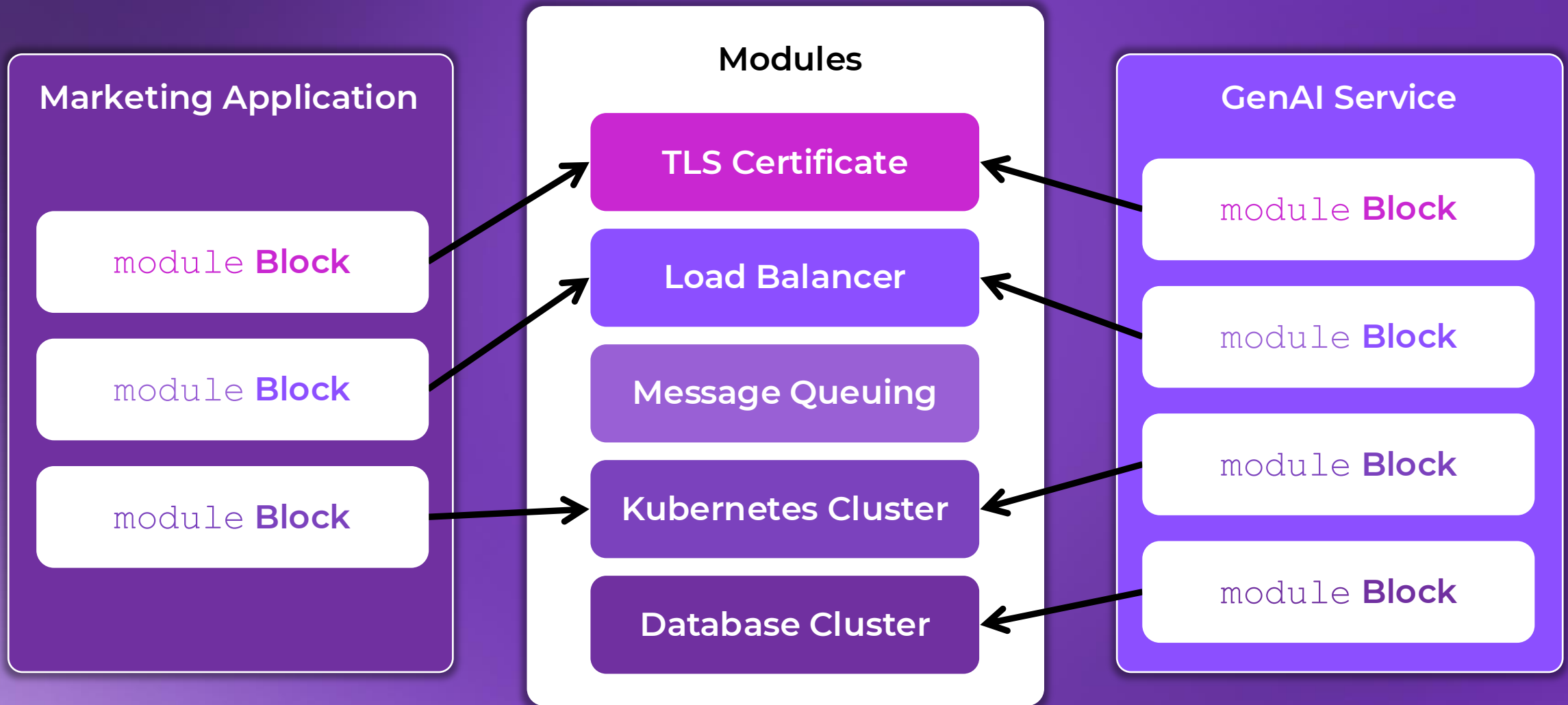
Consistent Patterns

Enforce naming conventions, security controls, and best practices across deployments.

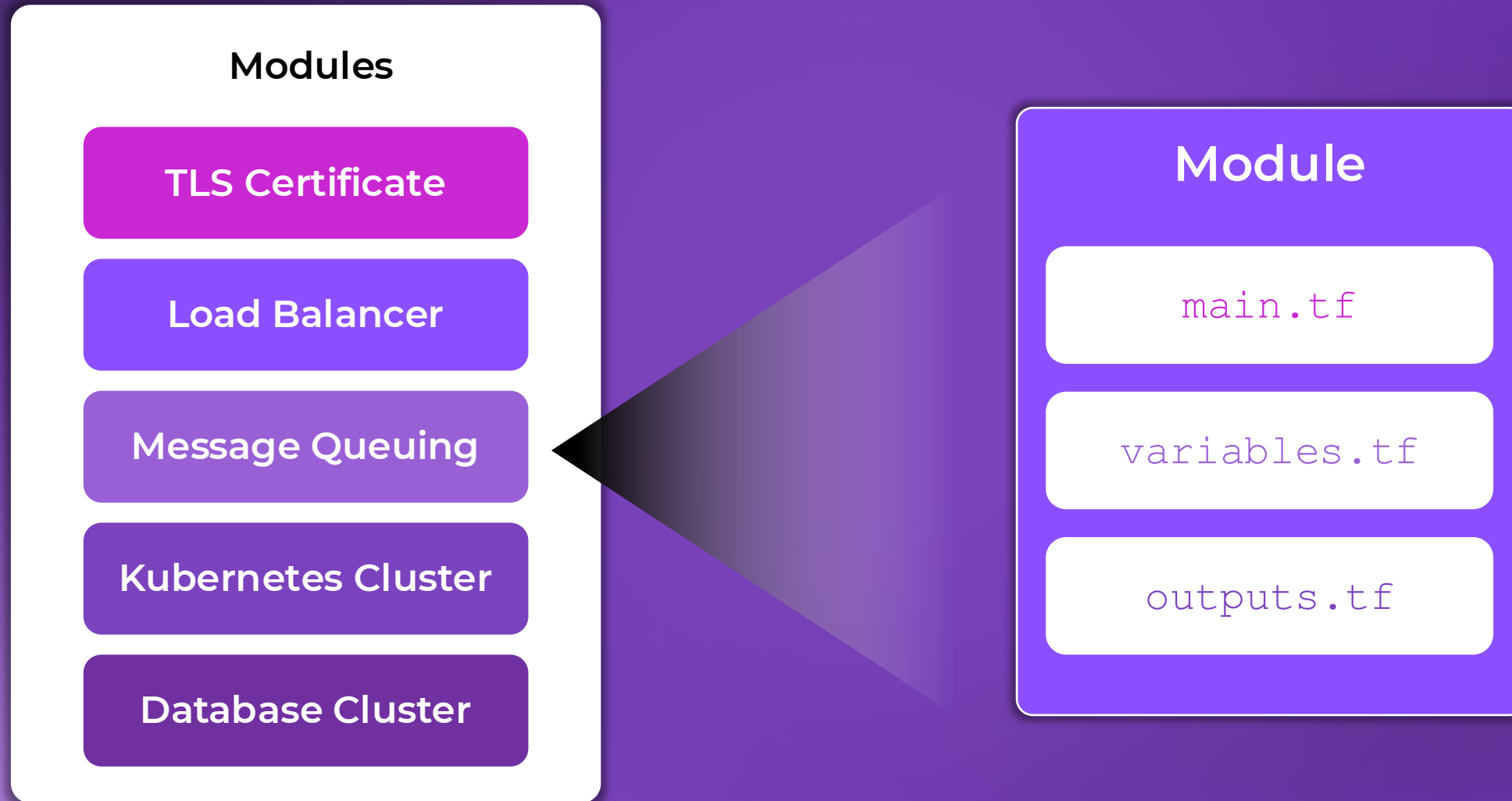
Terraform Building Blocks



Terraform Building Blocks



Terraform Building Blocks

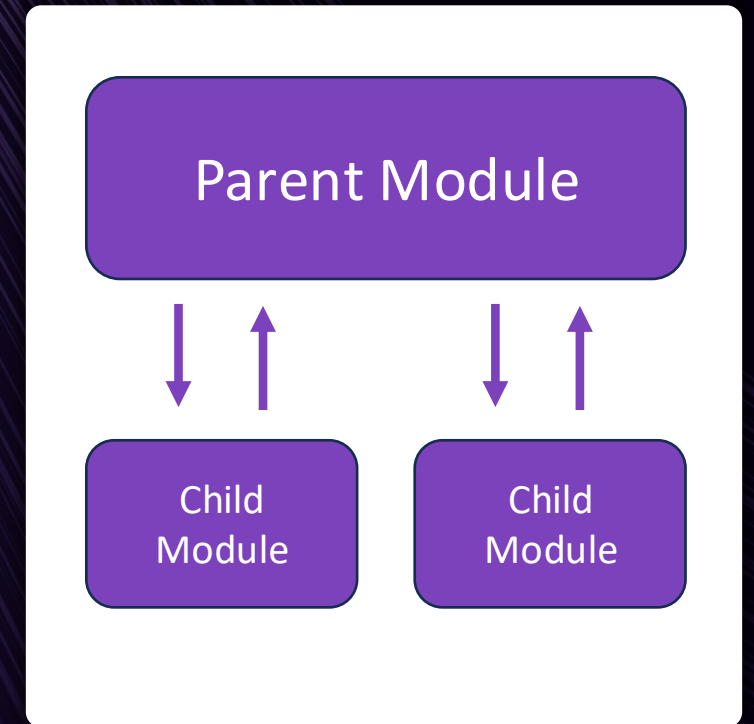


Module Block

Parent Module vs. Child Module



- ❧ A parent (calling) module references and configures other modules, while a child module is the reusable infrastructure component being called.
- ❧ Modules can be retrieved from a registry, a repo, or stored locally.





Parent/Root/Calling Module

```
1 module "vpc" {  
2   source = "terraform-aws-modules/vpc/aws"  
3  
4   name = "my-vpc"  
5   cidr = "10.0.0.0/16"  
6  
7   azs          = data.aws_availability_zone.east.name_suffix  
8   private_subnets = var.private_subnets  
9   public_subnets  = var.public_subnets  
10  
11   enable_nat_gateway = true  
12   enable_vpn_gateway = true  
13  
14   tags = {  
15     CreatedBy    = "terraform"  
16     Environment  = "prod"  
17   }  
18 }
```



Module Blocks Call Child Modules

Defines that this module is the calling module of the child module



Must Define Where the Module Lives

Path to local directory or remote module source to download from



Variables Defined in Root Module

Values must be passed to the module to be used

Very Simple Child Module



```
1  variable "cidr_block" {
2    type    = string
3    default = "10.0.0.0/16"
4  }
5
6  variable "name" {
7    type    = string
8    default = ""
9  }
10
11 resource "aws_vpc" "vpc" {
12   name      = var.name
13   cidr_block = var.cidr_block
14
15   tags = {
16     Name = var.name
17   }
18 }
19
20 output "vpc_id" {
21   value = aws_vpc.vpc.id
22 }
```



Modules Should Look Familiar

Modules are just regular Terraform configurations that are variablized



Must Define One or More Resources

The child module is where the resource blocks are defined



Child Modules Have Outputs

Outputs allow you to pass information from one module to another

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Root Module

```
1 variable "cidr_block" {
2   type    = string
3   default = "192.168.0.0/16"
4 }
5
6 module "prod_vpc" {
7   source = "terraform-aws-modules/vpc/aws"
8
9   name = "my-vpc"
10  cidr = var.cidr_block
11 }
12
13 output "vpc_id" {
14   module.prod_vpc.vpc_id
15 }
```

Values we want to pass to
our module

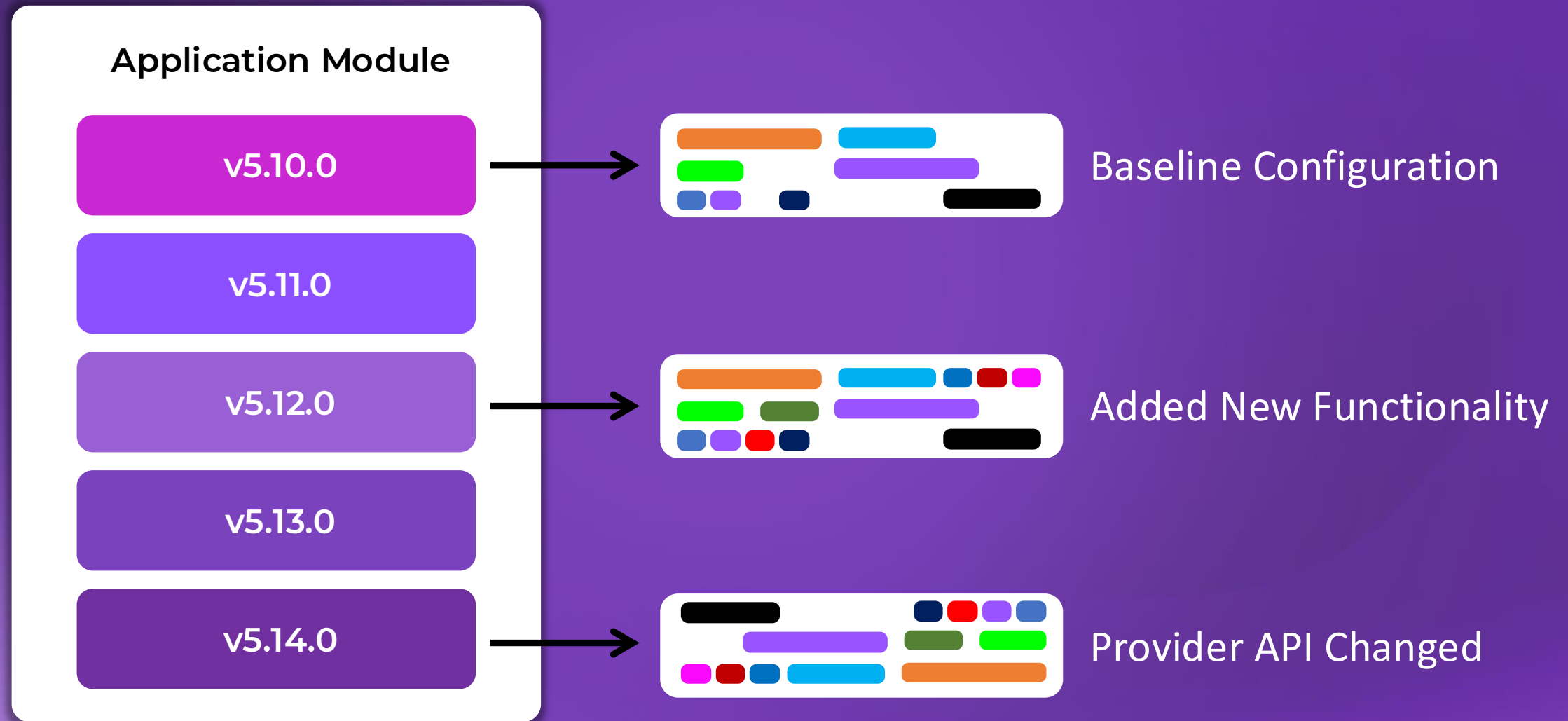
Export values to be accessed
by calling the module

Child Module

```
1 variable "cidr" {
2   type    = string
3   default = "10.0.0.0/16"
4 }
5
6 variable "name" {
7   type    = string
8   default = ""
9 }
10
11 resource "aws_vpc" "vpc" {
12   name      = var.name
13   cidr_block = var.cidr
14
15   tags = {
16     Name = var.name
17   }
18 }
19
20 output "vpc_id" {
21   value = aws_vpc.vpc.id
22 }
```



Module Versioning

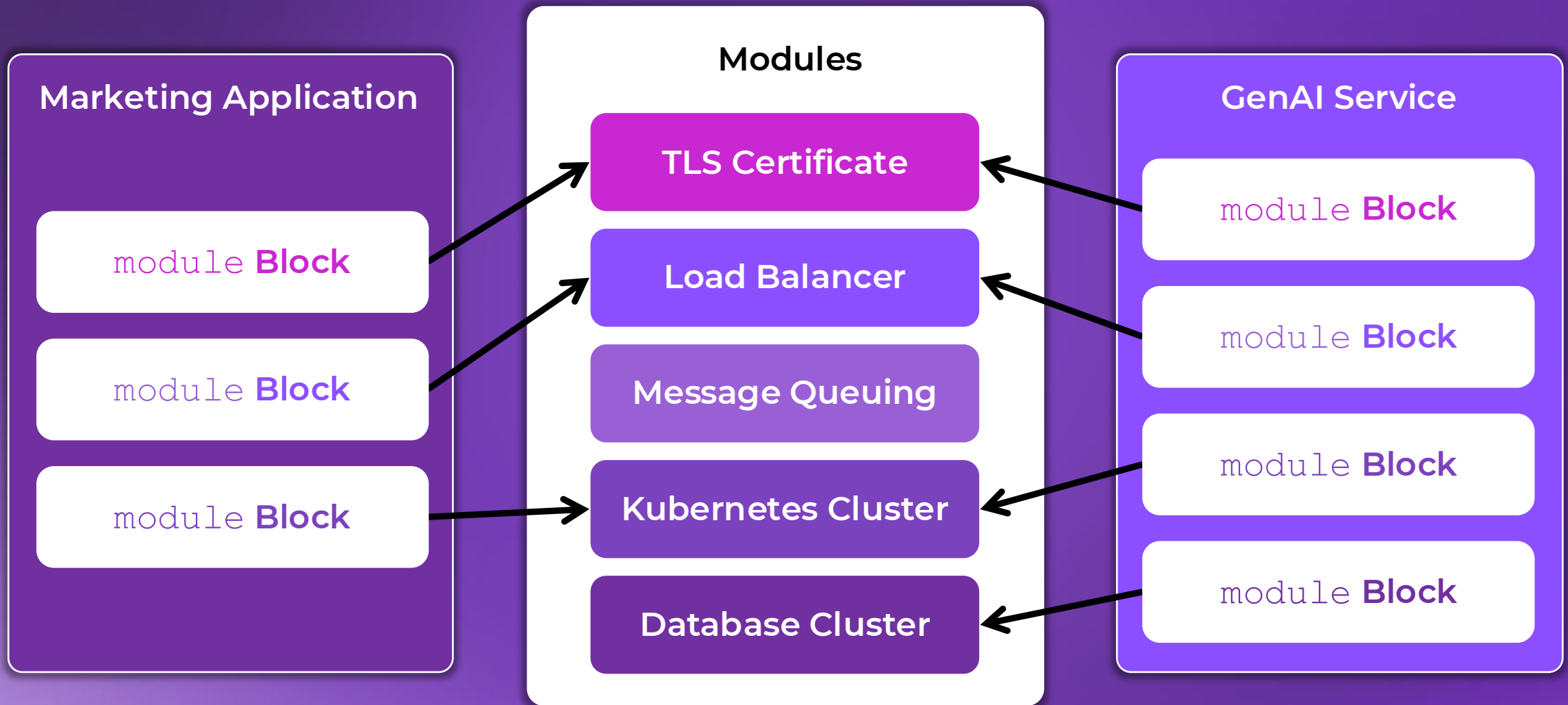


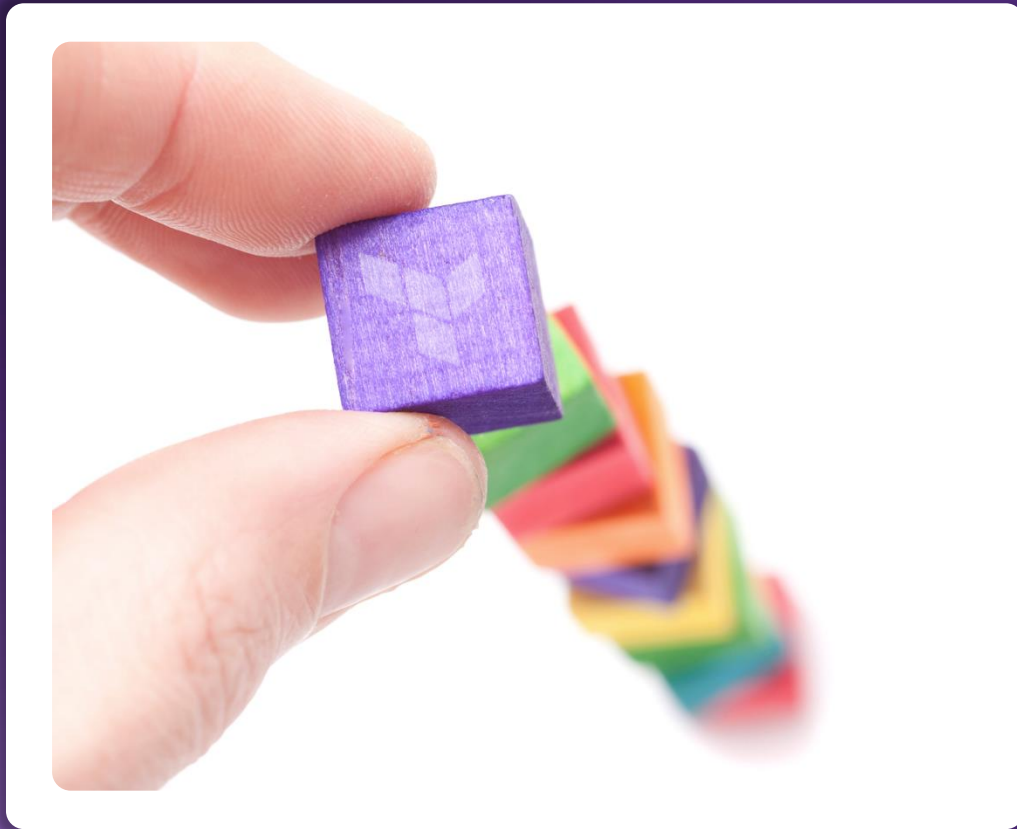
Pinning to a Specific Module Version

```
1 variable "cidr_block" {  
2   type    = string  
3   default = "192.168.0.0/16"  
4 }  
5  
6 module "prod_vpc" {  
7   source = "terraform-aws-modules/vpc/aws"  
8   version = "5.12.0"  
9  
10  name = "my-vpc"  
11  cidr = var.cidr_block  
12 }  
13  
14 output "vpc_id" {  
15   module.prod_vpc.vpc_id  
16 }
```



Terraform Building Blocks





Module Block

The **module** block allows you to reuse and organize Terraform configurations by calling pre-defined modules, which will simplify infrastructure management. Modules allow you to group resources into reusable components, such as a network or database module.

By passing variables to modules and using outputs, you can integrate them into your configuration for more dynamic and flexible deployments.



Module Block

Modules help you create maintainable and standardized infrastructure by packaging commonly used resources together.

● Code Reusability

Create standardized infrastructure patterns that can be used across multiple projects and teams.

● Simplified Management

Break complex infrastructure into smaller, manageable components that are easier to maintain and update.

● Consistent Deployment

Ensure infrastructure consistency by using tested, version-controlled module configurations across environments.



main.tf

```
module "vpc" {  
  source = "terraform-aws-modules/vpc/aws"  
  version = "4.0.1"  
  
  name = var.vpc_name  
  cidr = var.vpc_cidr_block  
  
  azs = ["us-west-2a", "us-west-2b"]  
  private_subnets = ["10.0.1.0/24"]  
  public_subnets  = ["10.0.101.0/24"]  
  
  enable_nat_gateway = true  
  single_nat_gateway = true  
}
```



Module Block

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main.tf

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module "vpc" {  
  source = "terraform-aws-modules/vpc/aws"  
  version = "4.0.1"  
  
  name = var.vpc_name  
  cidr = var.vpc_cidr_block  
  
  azs = ["us-west-2a", "us-west-2b"]  
  private_subnets = ["10.0.1.0/24"]  
  public_subnets  = ["10.0.101.0/24"]  
  
  enable_nat_gateway = true  
  single_nat_gateway = true  
}
```

Breaking Down the Module Block



```
main.tf

module "vpc" {
  source = "terraform-aws-modules/vpc/aws"
  version = "4.0.1"

  name = var.vpc_name
  cidr = var.vpc_cidr_block

  azs = ["us-west-2a", "us-west-2b"]
  private_subnets = ["10.0.1.0/24"]
  public_subnets  = ["10.0.101.0/24"]

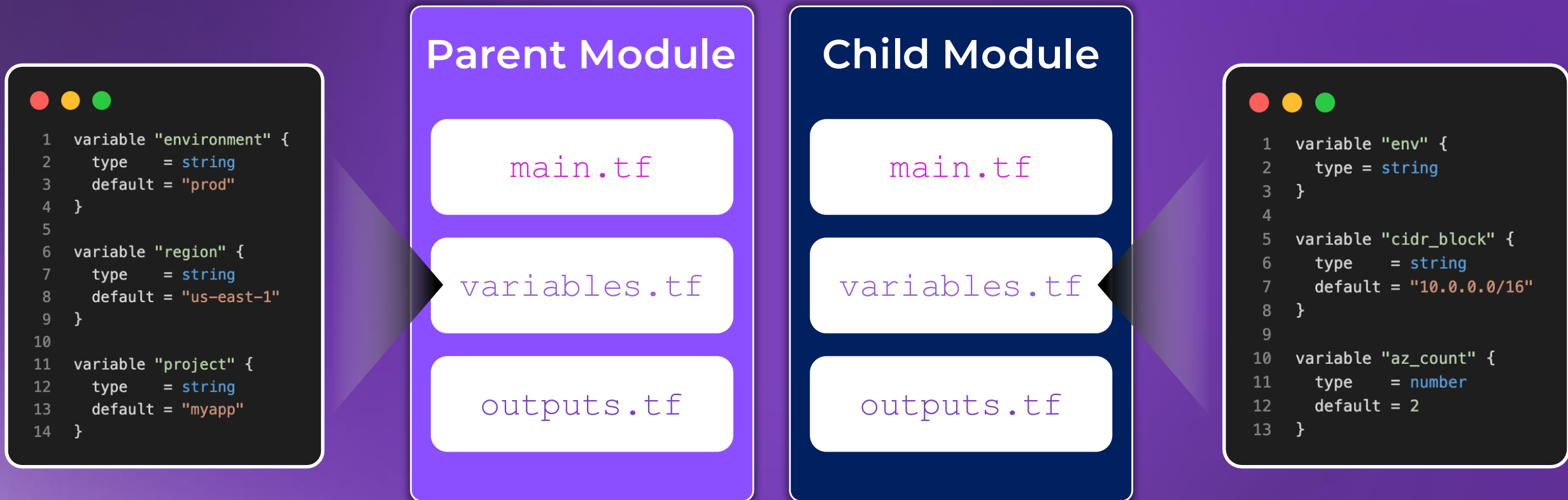
  enable_nat_gateway = true
  single_nat_gateway = true
}
```

Where the module is located to download

The version of the module to use

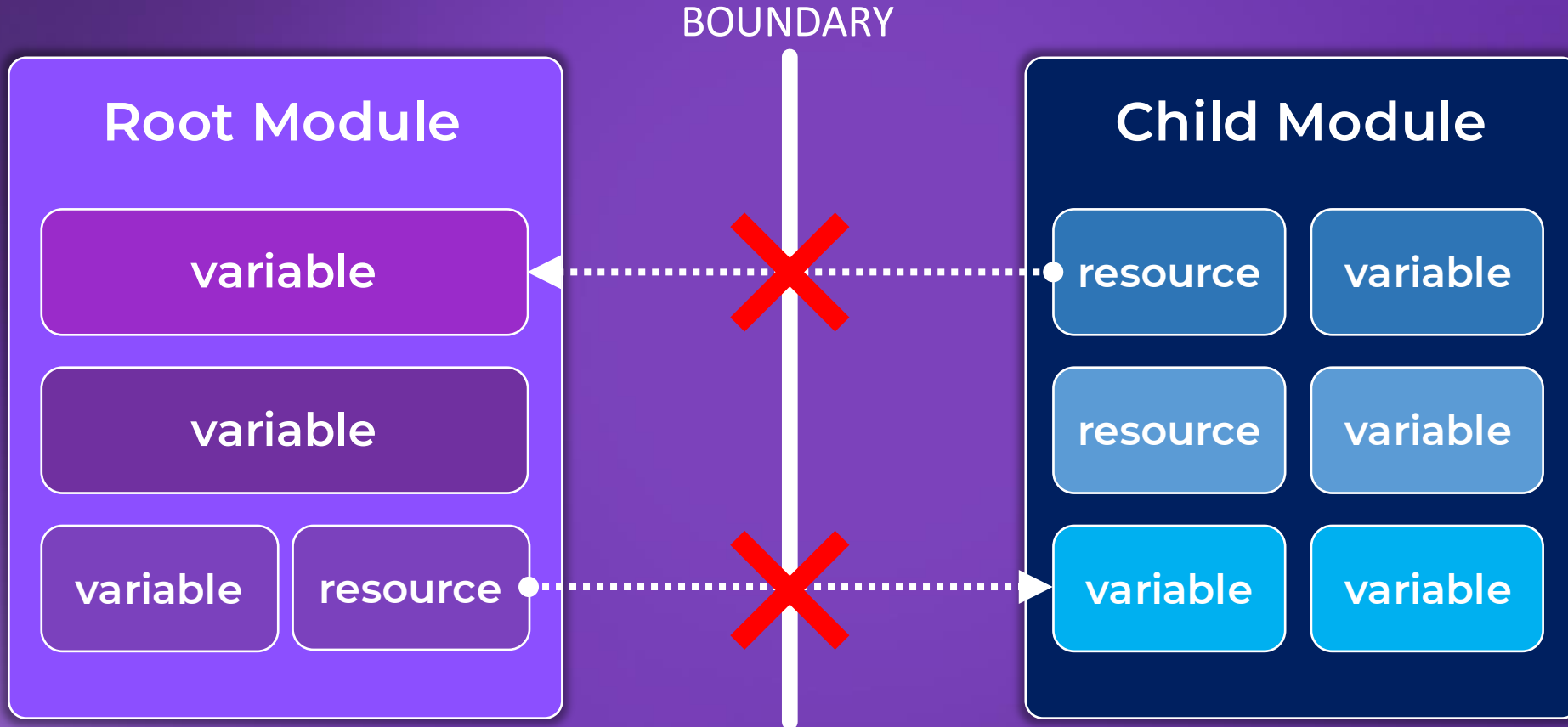
Local variables/values to pass into the module

Understanding Variable Scope in Modules



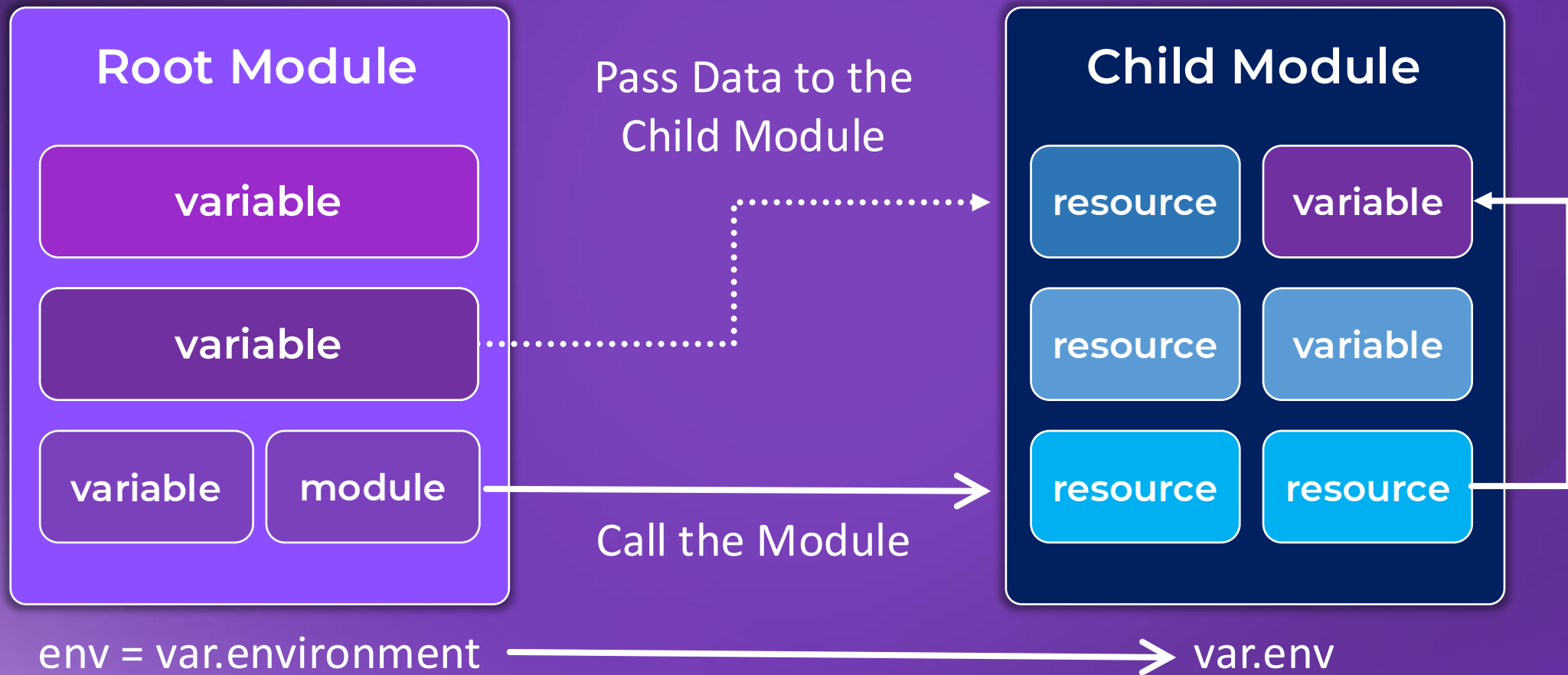
Both the parent and child module(s) may have variables defined

Understanding Variable Isolation



Each module has its own isolated variable scope

Passing Data to a Child Module



Passing Data to a Child Module



```
main.tf

module "vpc" {
  source = "terraform-aws-modules/vpc/aws"
  version = "4.0.1"

  name = var.vpc_name
  cidr = var.vpc_cidr_block

  azs = ["us-west-2a", "us-west-2b"]
  private_subnets = ["10.0.1.0/24"]
  public_subnets  = ["10.0.101.0/24"]

  enable_nat_gateway = true
  single_nat_gateway = true
}
```

Child Module

variable

variable

resource

variable

variable

resource

variable

variable

resource

variable

variable

resource

variable

variable

resource

Output Data Back to a Root Module

